

Statistical Overview of Current Antibody Peptide Sequences in OAS Database

5th semester bachelor in Biochemistry and Molecular Biology

ISA Project of Student

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Scientific motivation

The primary motivation for this project is to resolve the critical "coverage gap" in bottom-up proteomics. Current antibody identification relies on standard repositories like UniProt-SwissProt, which contain fewer than 40,000 sequences of human proteins - a negligible fraction of the billions of unique antibodies generated by the human immune system via V(D)J recombination alone. Consequently, disease-specific antibodies in complex biological samples often go undetected, limiting the diagnostic power of mass spectrometry (MS).

To address this, we focus on the comprehensive evaluation of the disease-specific peptide databases recently compiled from Next-Generation Sequencing (NGS) data (Trinh et al., 2024a). While these databases theoretically cover the missing "dark matter" of the antibody repertoire, they remain statistically unverified for proteomic use. This project aims to provide a rigorous statistical report of these resources, characterizing essential properties such as peptide recurrence (N_{patient}), antibody convergence (N_{antibody}), and isotype distributions across different disease states.

Introduction

Proteomics

Proteomics is the study of protein quantities in biological organisms. It concerns itself more with what is found, and how much rather than discovering new functions of individual proteins. The study of proteomics is therefore built on a framework of what we know and tries to use that framework on large data samples to gain an understanding of the system.

Proteomics is divided into bottom-up methods and top-down methods. Bottom-up proteomics starts with using tandem mass spectrometry of digested proteins. Generally, the workflow is to extract the proteins from the sample, then digest them. After that they are separated with liquid chromatography, and abundance is measured. Usually, the most abundant particles are then fragmented separately for identification. Then various bioinformatic tools are used to understand the data (McArdle & Menikou, 2021a, 2021b).

Top-down proteomics usually starts with the protein of interest and then finds it in the sample. It starts by accurately measuring the mass of the intact protein of interest. After having isolated the protein of interest, it is then fragmented to identify the individual amino acids and protein modifications. After that the data is compiled and analysed (Roberts et al., 2024).

Antibodies

Antibodies are proteins that are produced as part of the adaptive immune system. They are produced by B-cells and contain a large amount of variability. Antibodies are heterodimeric proteins consisting of two sets of light and heavy chains. Each chain has a variable region, light chains have 1 conserved region, and heavy chains have three conserved regions (Kretschmer et al., 2017). They are increasingly used in therapeutic treatments and for that reason it is of interest to understand the nature and distribution of antibodies to improve treatments. There is also an interest in the diversity of antibodies expressed in humans, and how they differ from case to case.

Variability of antibodies

Antibodies are randomly generated using a combination of two methods, V(D)J (somatic) recombination and junctional diversity. V(D)J recombination functions by combining different premade fragments of DNA for each variable region, this generates a basis for future modifications (Chi et al., 2020). It is a quick and easy way to fill out a lot of space with a good amount of diversity while taking up as little space in the genome as possible.

Meanwhile, junctional diversity occurs at the joints between the different segments. It is the addition or deletion of nucleotides in those areas (Janeway et al., 2001). The loss or gain of bases can affect the overall structure of the variable region which in turn makes it target something substantially different, this greatly increases the potential range of targets, than if it were purely the result of somatic recombination.

Class Switching

Antibodies come in different isotypes, the standard is Immunoglobulin M (IgM) which is produced by Naïve B-cells. It is the base for 3 of the other isotypes. The process of changing which isotype of antibody is being produced is called class switching (Liu et al., 2024). B-cells do not decide when to class switch, and what class to switch into. Class switching is regulated by the presence of cytokines, produced by Dendritic cells and/or T helper cells. Naïve b-cell respond to these cytokines and switch to producing the appropriate isotype of antibodies when triggered by t-helper cells (Stavnezer et al., 2008). The class switch itself is a rearrangement of the DNA of the class switching B-cell.

The different isotypes fulfil different functions in the immune system (Liu et al., 2024). This allows the immune system to make the antibodies fulfil different purposes as a part of the system with only a single progenitor, by class switching after the B-cell is activated there only

needs to be produced a single instance of B-cells that react to that target, which can then specialize against that threat.

Class switching is triggered by a B-cell reacting to antigens of the same protein fragment as a T-helper-cell, it also requires the connection of the B-cells CD40 receptor and the T-helper-cells CD40 ligand receptors. On top of the matching target requirements, class switching requires the presence of cytokines that are normally produced by the T-helper-cell. The nature of the cytokines will decide which class the B-cell switches to. This will trigger class switching and massive proliferation of the B-cell creating several B-cells with the same sequence in the variable region. During the proliferation, affinity maturation will occur to increase the affinity of the antibodies with the specific target, this will give several sequences with only minor variations (Song & Craft, 2019).

Isotypes

There are five different isotypes of antibodies in humans. The different isotypes have differing properties based on the organisation of their heavy chains constant area. Of the five isotypes, two of them have different subtypes.

IgM

IgM antibodies are the basic antibodies that are expressed on the surface of naïve B-cells. When class switching is triggered, the IgM is switched to either IgE, IgA, or IgG. In the blood IgM antibodies exist as pentamers. They are responsible for triggering the classical pathway of the complement system (Kretschmer et al., 2017).

IgM does not simply exist in order to class switch; they are part of a passive defence the body produces. They are the best activators of the complement system and seem to play a role in IgG mediated immunity. The sequence variation of IgM is the highest of any isotype, probably because they are produced passively without the need for a class switch. IgM and IgD antibodies are generated from alternative splicing of the same gene (Boes, 2000).

IgE

IgE antibodies are commonly associated with parasitic microbes and allergies. The conserved ends of the heavy chains have a high affinity with receptors found on mast cells, basophils, monocytes, and macrophages. The antibodies bind to Fc receptors. When the variable region binds to a recognized antigen it triggers cross-linking (Kretschmer et al., 2017).

The cross-linking uses the antibody to send a signal for the cells to degranulate. IgE isotype antibodies tendentially have a high affinity for the FcεRI receptor, therefore very few cross-links are needed to trigger the degranulation (Sutton et al., 2019).

IgA

IgA is the most produced type of antibody in the body, more IgA is produced than all the other antibody isotypes combined. There are two subtypes of IgA; IgA1 and IgA2. The differentiating factor is a hinge region on the heavy chains conserved part. IgA1 has a more flexible hinge region. These antibodies are mainly found two places in the body, the serum and the mucosal surfaces. They mainly exist as monomers in the serum, but in the mucosal membranes they are mostly dimers, sometimes even tetramers, these secreted IgA's are referred to as SIgA for secreted immunoglobulin A (de Sousa-Pereira & Woof, 2019).

In the mucus IgA antibodies hinder the accumulation of microbes preventing the formation of plaques and biofilms. They should not be able to activate the complement system as they lack the C1q binding site to do that, there have been cases that could indicate that they can trigger the alternative activation of the system (de Sousa-Pereira & Woof, 2019).

They are particularly effective at recruiting myeloid cells, but do not seem to activate the NK cells at the same time (Kretschmer et al., 2017). Considering the function of the NK cells being to deal with our own cells this could indicate that it is a preventative tool for rallying the innate immune system before the microbes begin invading, this would fit well with it not activating the complement system.

IgG

IgG antibodies are the most abundant of the isotypes in human serum. It comes in four different subtypes, IgG1 through IgG4. The four subtypes are each specialized in dealing with different things. All four subtypes can be produced at the same time, unlike if they were different isotypes (Vidarsson et al., 2014).

One of the differences between the different subtypes of IgG is the number of amino acids that make up the hinge region, with IgG3 being the longest while using more exons. The differences between the IgG's also affects their individual affinities and their lifespan. The different subtypes have different affinities and selectivity for the different gamma specific Fc receptors that they form an extension with, they therefore trigger different reactions. One of the main functions of IgG's is opsonization, being opsonized extends the lifespan of the antibody. (Kapur et al., 2014).

IgD

The function of IgD antibodies has been a mystery for a long time. While originally considered a relatively new addition to the antibody repertoire, it now appears to be an evolutionarily conserved antibody found from fish to humans. One of its enigmas, has been that a lot of its functions seem to overlap with the functions of IgM, but with it being so preserved across the animal kingdom the chance of it fulfilling vital functions is high. Maturation of B-cells is usually characterized by up-regulation of IgD production.

There is a possibility that IgD antibodies have a mediating effect on anergic B-cells, especially those that target self-antigens. It is theorised that this is a way to target microbes that express proteins that mimic host signatures to avoid being targeted by the immune system. It is known that IgD plays a role in the maturation of B-Cells, but the reason as to why this was evolutionarily tied together is still unknown. B-Cells have a special type of class switching using activation-induced cytidine deaminase (AID) that turns of their IgM antibodies making them pure IgD producers. These pure producers then migrate to body parts that interact with the outside world, with the noticeable exception of the intestines (Gutzeit et al., 2018).

With IgD being expressed together with other isotypes it makes the data murky. Are the entries registered as being IgD those cells that only expressed IgD or is there something else going on. IgD is encoded as an alternative splicing of antibodies, this allows it to use the same variable region while also fulfilling a different purpose, as it uses different exons it can also dodge deletion during class switch.

Affinity maturation

During sickness, B-cells are replicated in massive quantities. In order to improve the defence of the body, an evolutionary pressure is put on the B-cells. It forces them to continuously produce higher and higher affinity antibodies or die. While they are replicating, deliberate mutations are introduced into the variable region of the antibody's DNA (McHeyzer-Williams et al., 2011). This contributes to constant mutations that can potentially have better affinity for the antigens of the microbes. Before a new B-cell is allowed to replicate again it needs to be triggered by antigens, this causes the B-cells with a higher affinity to be favoured and thereby slowly enhances the average affinity of the antibodies of this lineage.

Observed Antibody Space

The observed antibody space is a database collection that was created in 2018, to combat the lack of organised data about antibodies. Its creation was prompted by the growing importance of antibodies in therapeutics. Currently it contains entries about 24 diseases, and healthy. To counter the problem that is the nonstandardization of antibody formats, the data is in the Minimal information about the Adaptive Immune Receptor Repertoire (MiAIRR) format (Olsen et al., 2022).

The difficulty of sequencing antibodies has led to alternative paths. The antibodies - that are recorded - in the observed antibody project are extracted via BCR-seq (Olsen et al., 2022). There is therefore a loss of information in the form of subtypes and actual antibody concentrations. The databases record the antibody sources, but not if there are differences between the productivity of these sources.

BCR sequencing

The databases of the OAS project are based on BCR (B-cell receptor) sequencing of patients. BCR uses either gDNA or cDNA to find out the variable sequence of the antibody. This can be done either in bulk or at single cell level. Each has its own downsides. Bulk sequencing is cheap but it mixes the nucleic acids in a pool, it is therefore not possible to pair the light and heavy chains whether gDNA or cDNA is used, as the heavy and light chains are encoded on different chromosomes. Single-cell sequencing solves the pairing problem by adding a barcode to each cell, but comes at a premium. Modern methods function for sequencing of under a million cells at once, and it is an expensive procedure (Kim & Park, 2019).

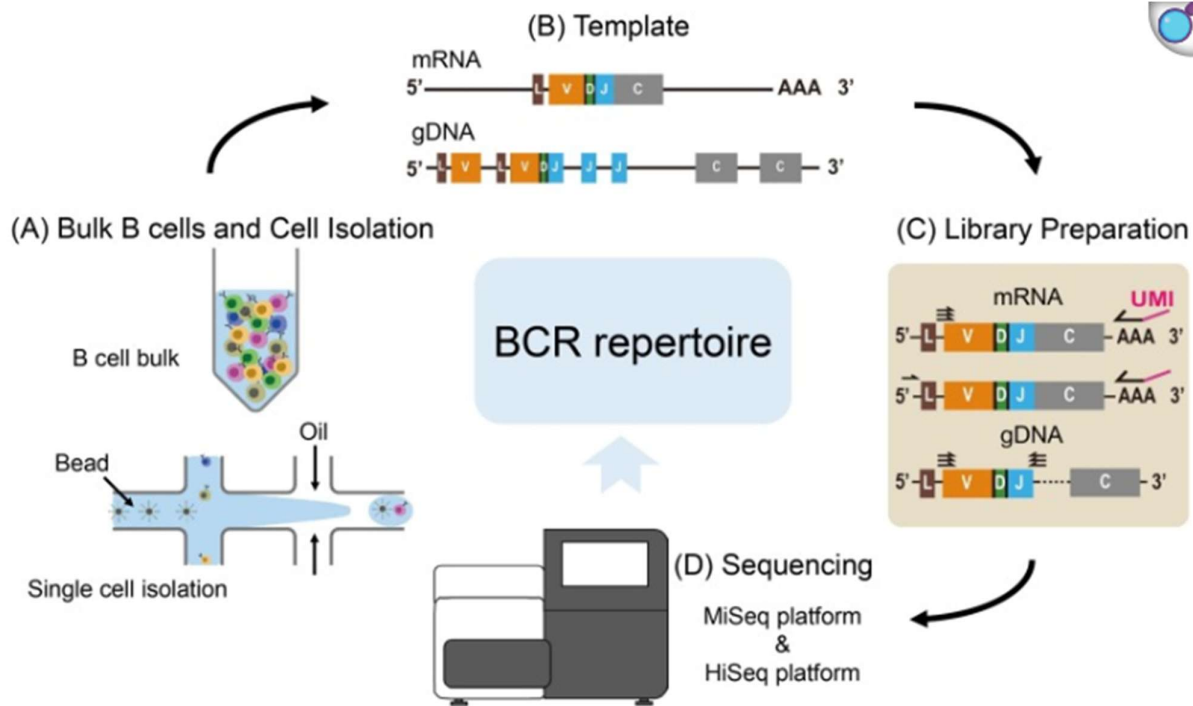


Figure 1: process of BCR sequencing from (Kim & Park, 2019).

The OAS uses the MiAIRR (Minimum Information about the Adaptive Immune Receptor Repertoire) format which is the current minimum standard for the notation of antibodies. Due to the wide range of existing processing pipelines direct comparison and standardization is complicated and a work in progress.

Proteomics of antibodies

The antibody data of the OAS is not from the proteins themselves; this is because the sequencing of antibodies is limited with modern techniques. Due to this there is a focus on sequencing the variable region as that is the region of interest - being the part that interacts

with antigens. This leads to a loss of information about the isotype, which decides the response that is triggered by the binding of the antigen (Olsen et al., 2022).

Data mining antibody sequences for database searching in bottom-up proteomics

“Data mining antibody sequences for database searching in bottom-up proteomics” is an article that seeks to analyse the underutilized databases in the OAS. It has cleaned and restructured the databases, the analyses in this paper were performed with the cleaned versions of the databases from. The aim of the paper was to improve the number of antibodies that can be recognized with bottom-up proteomics by analysing the theoretically available antibodies. This process will increase the rate of false positives, but those can be weeded out later in the process (Trinh et al., 2024b).

Project Description

Aim of the Project

The aim of the project is to continue the statistical analysis of genomic antibodies that has been previously done and create summary statistics of antibody classes like isotypes to identify patterns that can be linked to specific diseases.

Further work will investigate whether such classification is valid for data from blood proteomics mass spectrometry data, and thus whether the resulting isotype fingerprints can be used for disease diagnostics.

Methods and Experimental Design

The first step is to gather the previously created databases from https://github.com/trinhxt/SDU_Immunoinformatics, they are found on zenodo (Trinh et al., 2024a). Then the data will be used to investigate the antibodies at the sequence level and then create a summary of antibody isotype distribution for a per disease and patient. A through comparison of antibody isotype distributions between diseases will be performed to see if they are consistently different between diseases. The same procedure will be repeated on antibody peptide level including the inherent redundancies of multiple matches. If time permits, database searches of blood samples will determine whether the here calculated isotype distributions can be found in proteomics data.

The work will mostly be carried out via R scripting

Methods

Data Acquisition and Description

We utilized pre-compiled antibody peptide databases obtained from the Zenodo repository titled "Data mining antibody sequences for database searching in bottom-up proteomics" (Trinh et al., 2024, <https://doi.org/10.5281/zenodo.10561456>). These databases were originally constructed to address the limited antibody coverage in standard repositories like UniProt by leveraging next-generation sequencing of antibody gene repertoires. The dataset creators generated these sequences by annotating, translating, and numbering antibody gene sequences found in public repositories such as the Observed Antibody Space, subsequently digesting them virtually into tryptic peptides for mass spectrometry compatibility.

Data Processing and Statistical Analysis

We downloaded the disease-specific database files provided in *.*duckdb* format from the repository. Each database file corresponds to a specific disease condition (e.g., SARS-CoV-2, HIV, Celiac disease) and contains detailed metadata including peptide sequences, patient identifiers, and antibody isotypes.

Using these downloaded files, we performed a statistical summary to characterize the dataset. We quantified the distribution of data by calculating the total number of unique peptides and the number of antibodies associated with each disease category.

The databases were queried and duplicate entries were observed. Many of the duplicate entries were almost perfect duplicates - with the exception of their "N_antibody" values. The differing "N_antibody" could indicate that the duplicate entries were not actually duplicates, but overlapping information found at several timepoints, following this assumption they were therefore not discarded, but added. The duplicates were found via the SQL query to the Ebola database.

```
select Sequence, Patient, Isotype, BSource, BType, count(N_antibody) as N_entries,  
SUM(N_antibody) As total  
from DATDB  
Group by Sequence, Patient, Isotype, BSource, BType  
having count(N_antibody) > 1
```

It fetches every entry with a different; sequence, patient, isotype, B-cell source, and B-cell type. It then counts the amount of entries for each and the total sum of antibodies. It finally discards all entries that didn't have multiple entries. For the Ebola database the first six entries can be seen in **table 1**. The total amount of multientries for the Ebola database is 281704 rows.

Sequence	Patient	Isotype	BSource	BType	N_entries	total
DPGTGAGYLYHYMDVWGKTTVTSS	P92	IGHA	PBMC	Unsorted-B-Cells	3	12
GRPLYGEKSQGYWGQTLVTSS	P92	IGHA	PBMC	Unsorted-B-Cells	3	30
SDDTAVYYCARYMAVAHR	P92	IGHA	PBMC	Unsorted-B-Cells	2	2
YGEFPDYWGQGNLVTSS	P92	IGHA	PBMC	Unsorted-B-Cells	2	2
GLEWVSLISWDNDRTYYADSVK	P92	IGHA	PBMC	Unsorted-B-Cells	2	4
AEDTAVYYCAKDLFVEGSGSYWSTSK	P92	IGHA	PBMC	Unsorted-B-Cells	2	4

Table 1: table showing 6 repeat entries for the Ebola database. There are a total of 281 704 entries that are repeated at least twice.

After understanding the data structure several duckdb queries were used to get information about the data, this information was then graphed.

Results

Overview - Some basic house cleaning to keep things consistent

Referring to the data

Many diseases are getting increasingly complicated and long names, that makes abbreviations increasingly common. With a lack of standardisation in abbreviations this report will refer to diseases based on the following table. The “Full Disease Name” is the official full name of the disease, the “Name in data” is the name that is used in the OAS, and the abbreviation is the way I refer to it when there is a lack of space.

Index	Full Disease Name	Name in data	Abbreviation
1	Severe Accute Respiratory syndrom Coronavirus 2	SARS-CoV-2	SARS-CoV-2
2	Human Immunodeficiency Virus	HIV	HIV
3	Co-infection of Cytomegalovirus and Epstein-Barr Virus	CMV-EBV	CMV-EBV
4	Celiac Disease	Celiac	CD
5	Allergy without Specific Immunotherapy	Allergy-NoSIT	Allergy-NoSIT
6	Ebola Virus	Ebola	EBOV
7	Multiple Sclerosis	MS	MS
8	Systemic Lupus Erythematosus	SLE	SLE
9	Tonsillitis	Tonsillitis	Tons.
10	Obstructive Sleep Apnea	Obstructive Sleep Apnea	OSA
11	Tonsillitis Obstructive Sleep Apnea	Tonsillitis Obstructive Sleep Apnea	Tons./OSA
12	Asthma	Asthma	A
13	Epstein-Barr Virus	EBV	EBV
14	Hepatitis C Virus	HCV	HCV
15	Acetylcholine Receptor–positive Myasthenia Gravis	AChR-MG	AChR-MG

16	Light Chain Amyloidosis	Light Chain Amyloidosis	AL Amyloidosis
17	Muscle-Specific Kinase–positive Myasthenia Gravis	MuSK-MG	MuSK-MG
18	Cytomegalovirus	CMV	CMV
19	Allergy with Specific Immunotherapy	Allergy-SIT	Allergy-SIT
20	Dengue Virus	Dengue	DENV
21	Chronic Lymphocytic Leukemia	CLL	CLL
22	Seasonal Allergic Rhinitis	Allergic Rhinitis in Season	ARIS
23	Non Dengue Febrile Illness	Non Dengue Febrile Illness	NDFI
24	Perennial Allergic Rhinitis	Allergic Rhinitis Out of Season	AROS

Table 2: An overview with the full names of the diseases, the registered name in OAS, and an abbreviation for use when later in some figures, depending on space.

Metadata

Discrepancy

The only problem found in the metadata was that. According to the metadata the number of patients in our largest database, the SARS_COV_2 is 180, but the script told us that there were only 179 distinct patients.

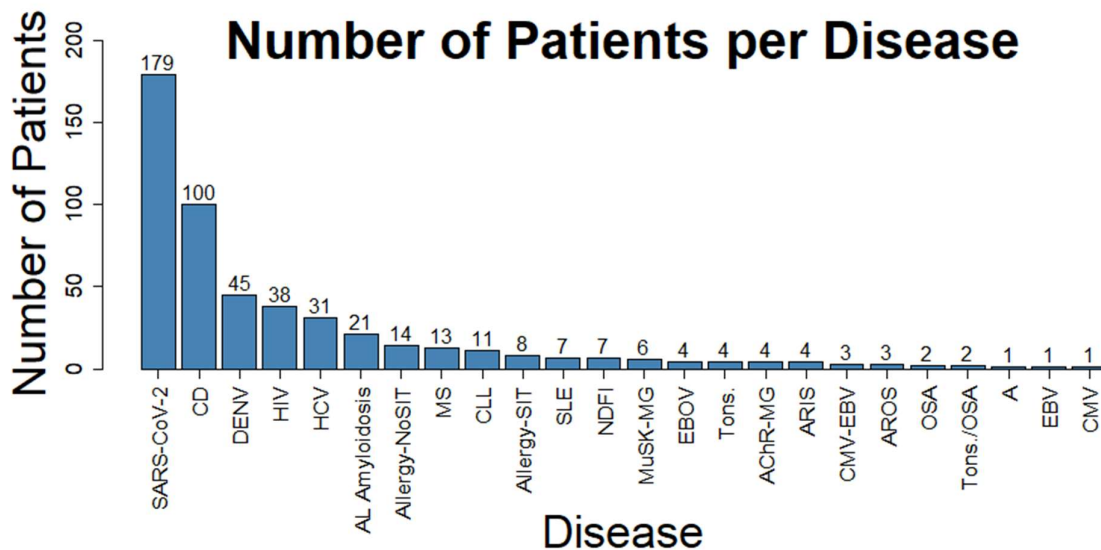
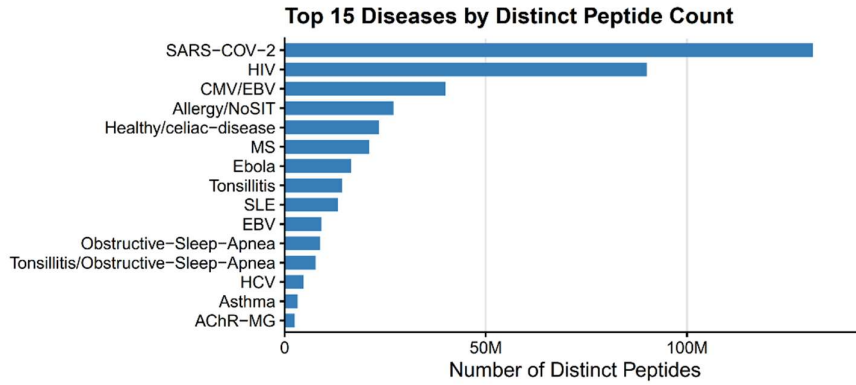


Figure 2: the number of patients per dataset, sorted in descending order.

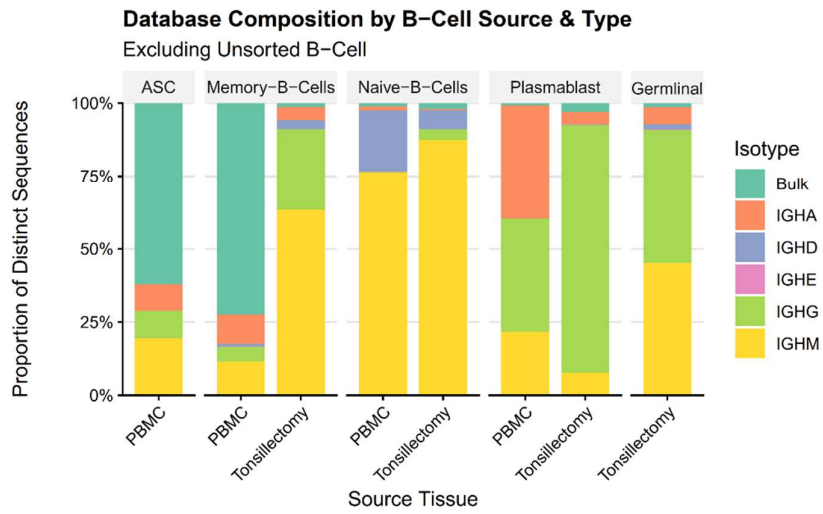
Quite luckily the discrepancy occurs in the SARS_CoV_2 data which has by far the most patients, so the loss of one should not affect any outcomes too much.

Quick overview

A



B



C

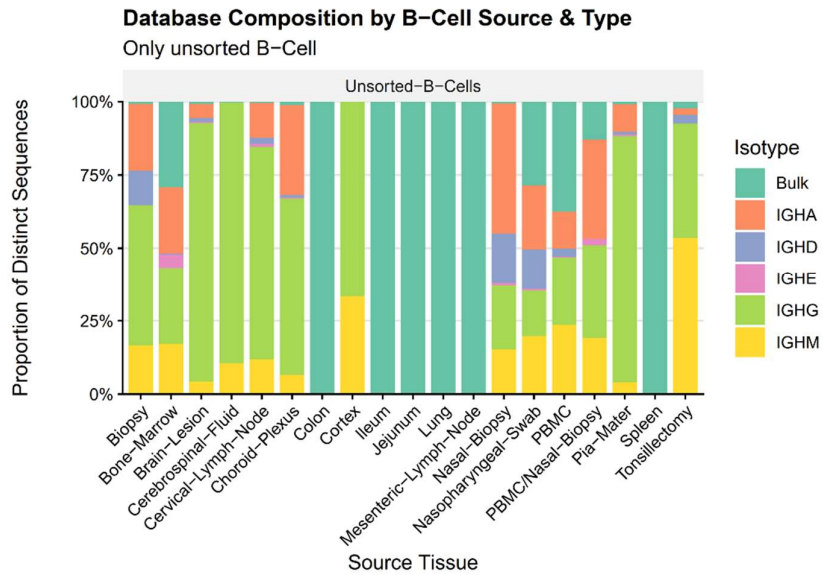


Figure 3: Database Composition and Demographics. (A) Bar chart displaying the top 15 diseases by total count of distinct peptides. (B) Stratification of distinct peptide sequences by B-cell source (tissue), cell type, and isotype. (B1) Defined B-cell subsets. (B2) Unsorted B-cells.

Global disease peptide count and diversity of origin, and isotype distribution.

Figure 3A shows the top 15 diseases ranked by distinct peptide count, with viral infections dominating the database. SARS-CoV-2, HIV and CMV/EBV contain the largest peptide repertoires. This reflects extensive sequencing efforts and the broad humoral responses these diseases trigger. While the sampling depth of peptides captured for different disease in the database varies, it does cover several areas, such as; autoimmune, inflammatory, allergic, and healthy immune states. It therefore enables comparative analysis of different immune contexts, of antibody-peptide analysis.

B-cell source, cell type, and isotype composition

Figures 3B and 3C summarize database composition across B-cell sources, cell types, and isotypes. Defined B-cell populations display coherent isotype patterns, with naïve and memory B cells enriched for IgM and IgG, and tissue-derived and plasmablast populations dominated by class-switched isotypes such as IgG and IgA. Unsorted B-cell samples show greater heterogeneity across tissues, reflecting mixed cellular composition and study-specific sampling. Overall, these patterns indicate that the database preserves expected features of B-cell biology despite its scale and heterogeneity.

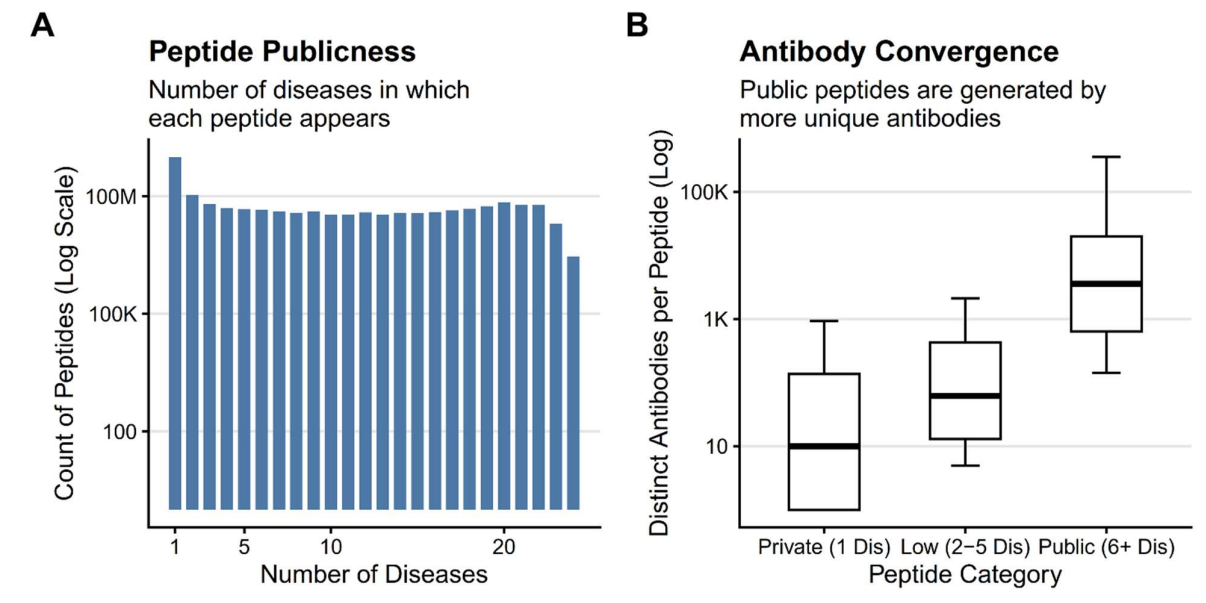


Figure 4: **(A)** Number of diseases (Dis) in which each peptide appears. **(B)** Public peptides shared across more diseases are generated by larger numbers of distinct antibodies, indicating increased antibody convergence.

Peptide recurrence across diseases

Figure 4A shows the recurrence of peptides across diseases, the publicness of a peptide. Most peptides show up in mainly a single disease, with the amount of peptides that are recurrent in 2 through 22 diseases being relatively stable. While cumulatively less than the ones that show up in only a single disease these recurrent peptides make up a significant portion of the data, due to a single recurrent peptide being able to appear for each disease that it is present in. This indicates a recurrence of sequence motifs rather than strictly disease-specific peptides.

Antibody convergence on public peptides

Figure 4B demonstrates a potential correlation between the publicity of the peptide and the amount of sequenced B-cells that produce that peptide. More public peptides are generally produced by more individual b cells than their more private counterparts. This could indicate that more public peptides arise from convergent scenarios. With antibodies being generated by first somatic recombination there is good reason to believe that there will be many antibodies that are generated with a highly similar structure. The antibodies will then undergo affinity maturation to match their specific disease which would create an antibody that is highly specialised in dealing with that disease. While the database has divided the patients based on a single disease, that is little doubt that there are many other diseases that these patients have had during their lifetime that overlap. With that there would be reason to believe that public peptides are related to these diseases that they share.

Bulk databases

Some of the patients only have bulk registered as the isotype for their antibodies, and the sequence part of the data only records the variable region. Considering patient count of some of the databases this could quickly make them useless for anything to do with isotyping.

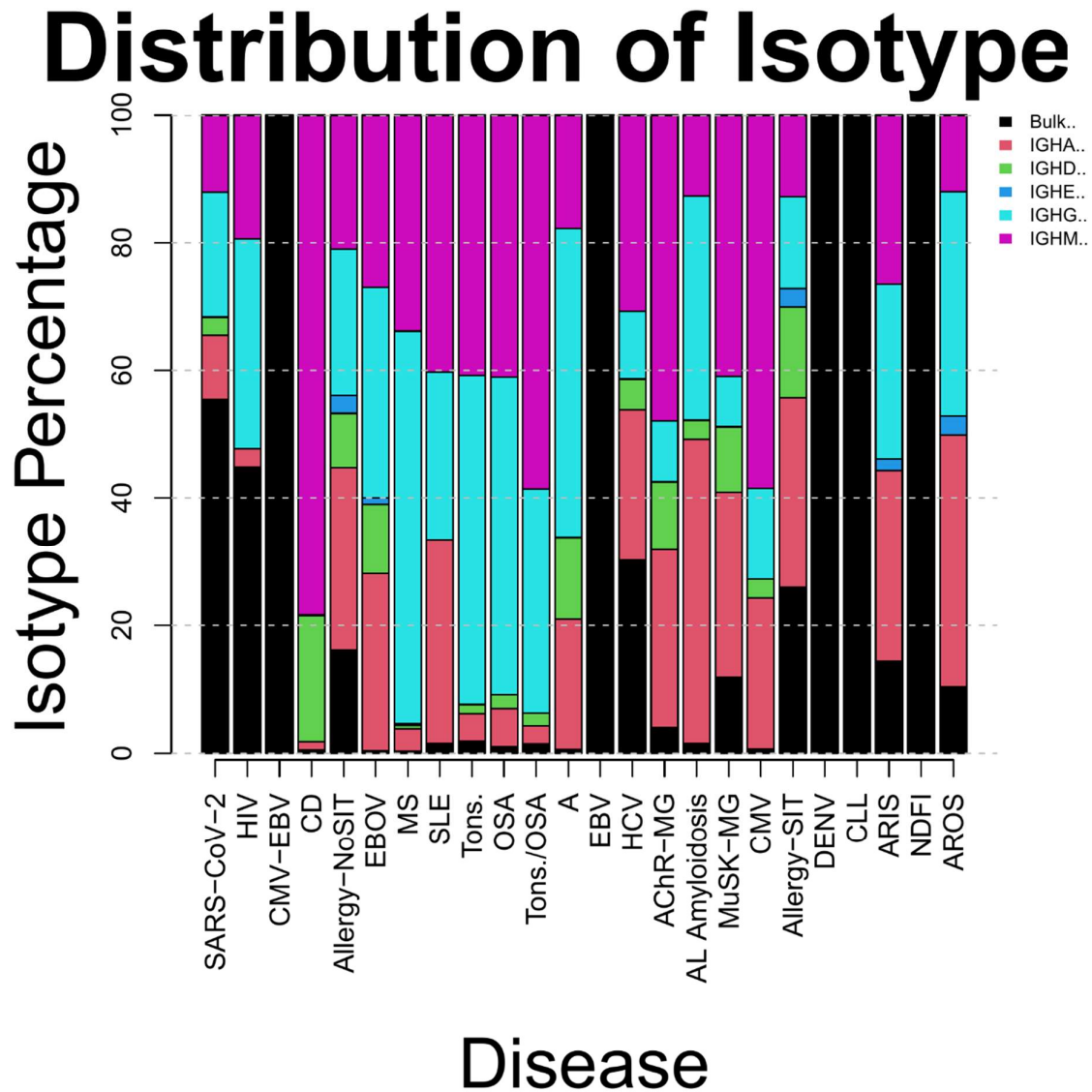


Figure 5: Percentage isotype distribution, all entries.

Isotype distribution

Based on the theory about IgA antibodies, they should be the most common isotype of antibody in the body. **Figure 5** shows that while IgA is present in all datasets, it is not the overwhelming majority of B-cells. This is probably a result of the OAS using BCR sequencing for their count. If the data is reliable and true, it would indicate that the comparatively larger production of IgA, is covered by a similar amount of B-cells of the IgA class as the production of other isotypes. For IgA to be the most abundant protein isotype in this situation it is

necessary for the IgA cells to individually be producing a lot more antibodies than their counterparts. This would fit well with the thought that IgA antibodies are a preventative measure rather than a reactive measure.

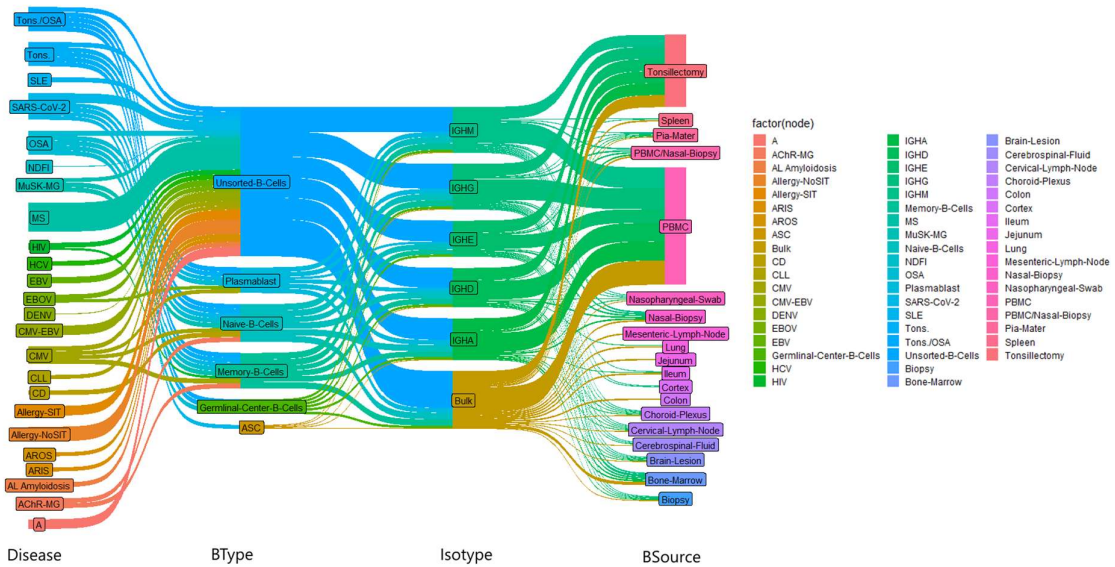


Figure 6: A Sankey diagram showing the relationship between Isotype and B-cell source and B-cell type.

Figure 6 attempts to map the isotypes to where they are taken from. An interesting thing to note is that there are naïve B-cells of all isotypes. According to the theory that should not be the case, for a B-cell to class switch it needs to be triggered by a t-helper cell. Interacting with a T-helper cell would require that the B-cells are in an active state, as in not being naïve. This could be a quirk of the data, or an interesting find

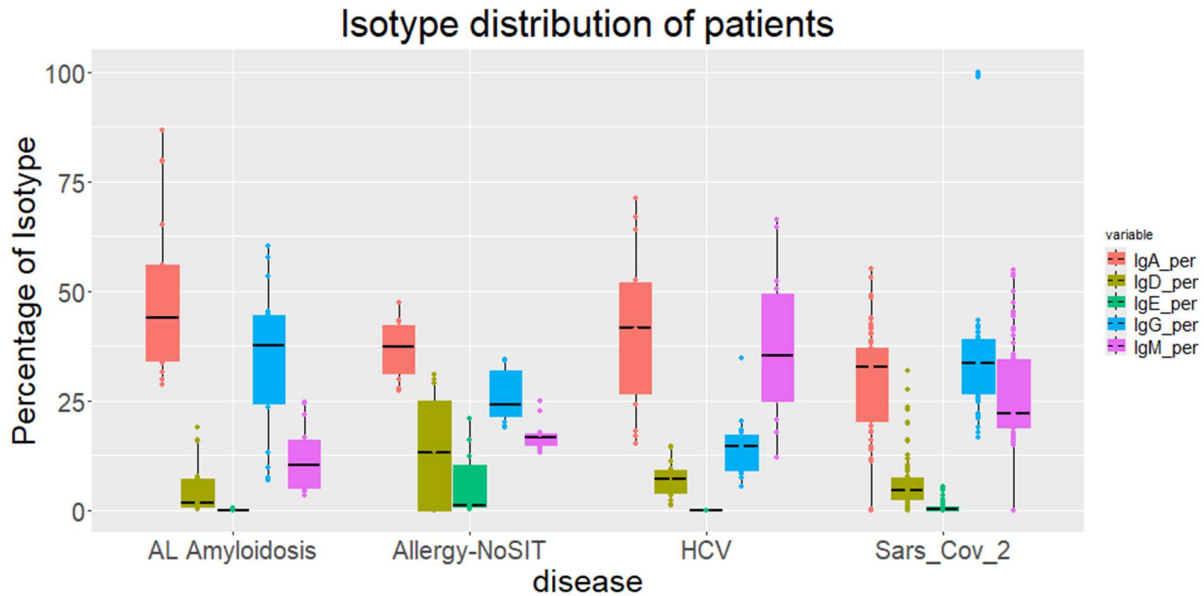


Figure 7: the distribution of patients isotype percentage for 4 major databases. Each dot is a patient, each patient is represented 5 times in their disease, one for each isotype. The y axis is the percentage of B-cells in that patient of that isotype, with Bulk removed under the assumption that it would represent the same distribution to make the diseases comparable.

Figure 7 is a boxplot comparing the distribution of isotypes for each patient in 4 databases, the databases were chosen based on which had the most patients left after removing all the patients with an missing entries, as those missing entries would skew the percentage distribution of the isotypes that were not missing. Of potential interest are the low levels of IgE in all but the Allergy database, this can probably be explained by allergy being a hyperimmune reaction where the immune system targets harmless antigens. When properly functioning IgEs are produced as a response to helminths, and allergy has a similar response pattern to that.

The IgD levels found in allergy cannot be explained the same way, IgD is produced by most B-cells along with whichever other isotype they normally produce. With the assumption that the IgD entries in the data are those that purely produce IgD it may be that the production of high levels of IgE signals that the entrants to the body need to be better protected. If that were the case it would rather be expected that IgA production would increase instead of IgD as that seems to be the main antibody in mucos. As a counterpoint to that, IgE has evolutionarily been produced to counter larger threats, where the activation of macrophages and neutrophils hasn't been enough, maybe IgD fills some special role there.

For all four diseases IgA seems to be one of the more common types to meet. This could be because it is a preventative isotype. The only disease where IgA seems to be less represented

in is Allergy, and that may be for the reason discussed above as you generally get diminishing returns on preventatives.

Conclusion

The lack of proper data on the area makes it hard to say anything definite, but there is clear proof that some diseases produce different isotype fingerprints when people have them. There is not yet enough data to draw a perfect picture for any one disease, but there is a visible difference between the likes of Sars-Covid-2 and Allergy, as should be expected. With further research it is likely that it will be possible to sort diseases into categories based on their isotype fingerprint. If the categorisation of diseases can be done properly it may be possible to use that fingerprint in the future to gain an understanding of the nature of new diseases. To understand if they conform with what we expect, or cause the body to react in a weird way.

While there is a wealth of information about antibodies for some diseases, there is a lack of information for most of the others. If there was more information available it might be possible to sort diseases into categorie, but currently there is not enough information available to do that. But if the isotype fingerprint could be gleamed, it would make it possible to understand how the body fights that disease and allow us to help it there. It would also allow us to prevent much of the harm a disease can do as much of the damage that comes from diseases is the body accepting some damage now to remove the problem.

The information that is available has previously been used to perform proteomic and that is probably the best use for the datasets in their current state. Potential future work could focus on trying to expand the available data or use other parts of it. A potential use case for the sequence registration is the creation of phylogenetic trees to document the maturation of antibodies over time, and if that is combined with the mapping of isotype fingerprints, perhaps some new knowledge or patterns can be gleamed. Due to the maturation happening one base at a time, it should theoretically be possible to make a complete map of the transformation that tracks the change one amino acid at a time.

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